

Operating Systems: Three Easy Pieces

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0.1 Introduction to Operating Systems

- Crux of the problem: How does the operating system virtualize resources?
 - What mechanisms and policies are implemented by the OS to attain virtualization?
 - How does the OS do this efficiently?
 - What hardware support is needed?

Definition 0.1 (Virtualization). The process in which the operating system takes a physical resource and abstracts it into an easier-to-use virtual form.

Definition 0.2 (System call). An interface or API exposed to the user to tell the OS what to do. A typical OS will export on the order of hundreds of system calls (syscall)

- The OS is also responsible for managing resources, each of the CPU, memory, and disk.

```
1 int main(int argc, char *argv[]) {  
2     if (argc != 2) {  
3         fprintf(stderr, "usage: _cpu_<string>\n");  
4         exit(1);  
5     }  
6     char *str = argv[1];  
7     for (;;) {  
8         Spin(1);  
9         printf("%s\n", str);  
10    }  
11    return 0;  
12 }
```

- The above code will simply just print the string that the user passed, forever.
- Interestingly though, even on a single core system we can run this program in parallel and all programs will run at the same time.
- The OS will virtualize each physical CPU into a seemingly infinite number of virtual CPUs, giving the illusion as if there are more than one.
- The OS policies defines the behaviour of the OS, including things like what programs get priority and what actions are permitted.
- A similar idea is applied with memory: memory is just an array of bytes.
- To read/write to memory, one needs to supply the address of whatever you want to access.
- Consider a program that allocates some memory, prints the address of the memory `p` and writes the number 0 into the first slot of the allocated memory. Finally it loops and increments the value stored at the address held in `p`.
- Similar to the example from before, if we run multiple instances of this program, then each instance will have allocated memory at the exact same address and each are updating the value at that address independently.
- This is an example of the OS virtualizing memory; each process has it's own private virtual address space.
- It is the OS's responsibility to map the virtual memory of each process to some portion of physical memory.

- Persistence: The OS is also responsible for ensuring that data that is stored in a volatile manner (such as DRAM) is not lost when power is lost or the system crashes. This is done through the usage of input/output, or I/O devices.
- Common IO devices in modern computers are hard drives or SSDs.
- Unlike the abstractions the OS performs on the physical CPU(s) and memory, there is no virtualization performed for each application.
- It is assumed that the user wants to share the contents of a file; one application may be used to create and edit a file and another to compile.
- The software that is responsible for managing and storing files on the disk is called the file system.

4 The Abstraction: The Process

- The crux of the problem: How to provide the illusion of many CPUs? In a system where there are only a few physical CPUs available, how can the OS provide the illusion of a nearly-endless supply of CPUs?
- The basic idea is simple: On a given CPU, the OS will run one process, then stop it to run another process, and so on. This is known as **Time sharing**.
- Using this the OS can provide the illusion that one CPU is running multiple processes, at the cost of slower performance the more processes need to be shared on one CPU.
- Mechanisms: low-level methods or protocols that implement functionality.
- Policies: higher-level algorithms that are responsible for making some kind of decision within an OS.
- A process at a given time can affect the memory that is reserved for it (its address space), as well as registers.
- Special registers include the Program Counter/Instruction pointer (PC/IP), which tells what the next instruction to be executed is and a stack pointer is used to manage the stack for function parameters, local variables and return values.
- Programs may also affect persistent storage devices as well.
- The OS must expose the following APIs for processes:
 1. Create: An OS must be able to create a new process.
 2. Destroy: An OS must be able to destroy an existing process.
 3. Wait: It is sometimes useful to wait for a specific process to stop
 4. Misc. control: Some way to temporarily halt and resume a process
 5. Status: Read-only APIs that give status info or some other pieces of metadata on a process.

4.1 Process Creation

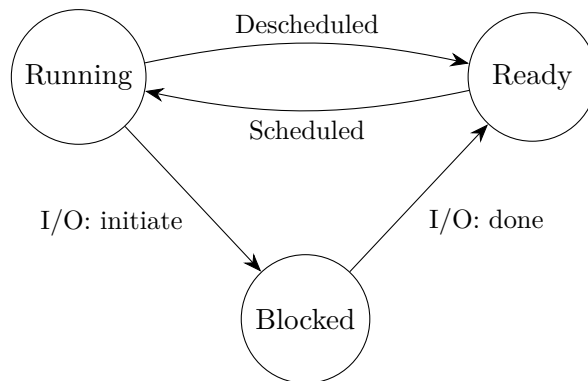
- The first thing the OS does is load the program code and any static data into memory (the address space of the process)
- This is done lazily; the OS will only fetch pieces of code or data that is necessary (opposed to eagerly). This is done to further support paging and swapping.
- The OS then allocates memory on the program's stack as well as the heap.
- It will then perform any remaining initialization tasks, such as those pertaining to IO. The OS then transfers control of the CPU to the newly-created process.

4.2 Process States

In a simplified view, a process can be in one of three states:

1. Running: The process is currently being executed.
2. Ready: A process is ready to run but is not currently being executed for some reason from the OS.

3. Blocked: A process has performed some operation in which it needs to wait for in order to proceed. An example is an IO request, in which the process needs to wait for the data to be fetched and is thus blocked; in the meantime some other process is free to use the processor.



Below is an example for how two processes may transition through some of the states as shown in the diagram above:

Time	Process ₀	Process ₁	Notes
1	Running	Ready	
2	Running	Ready	
3	Running	Ready	Process ₀ initiates I/O
4	Blocked	Running	Process ₀ is blocked,
5	Blocked	Running	so Process ₁ runs
6	Blocked	Running	
7	Ready	Running	I/O done
8	Ready	Running	Process ₁ now done
9	Running	–	
10	Running	–	Process ₀ now done

4.3 Data Structures

- The process list will contain all processes that are ready and some extra information for what processes are currently running.
- The OS must also keep track of blocked processes.
- The register context will hold the contents of the registers of a stopped process. The registers of a stopped process are written to memory, and are restored when the process resumes.

Below is an example of some data structures used in the xv6 kernel:

```

1 // the registers will save and restore to stop and restart a process
2 struct context {
3     int eip;
4     int esp;
5     int ebx;
6     int ecx;
7     int edx;
8     int esi;
9     int edi;
10    int edp;
11 };
12
13 // the possible states a process can be in:
14 enum proc_state { UNUSED, EMBRYO, SLEEPING, RUNNABLE, RUNNING, ZOMBIE
15     };
16
17 // the information xv6 tracks about each process
18 struct proc {
19     char *mem; //start of process memory
20     uint sz;
21     char *kstack; // bottom of kernel stack for this process
22     enum proc_state state;
23     int pid;
24     struct proc *parent;
25     void *chan; // !zero -> sleeping on chan
26     int killed; // !zero -> killed
27     struct file *ofile[NOFILE]; // open files
28     struct inode *cwd;
29     struct context context;
30     struct trapframe *tf;
31 }

```

As seen above there are more states that a process can be in than the three listed;

- An initial state for when it is being created
- A final state where it has exited but has not yet been cleaned up (zombie)

5 Interlude: Process API

5.1 The `fork()` System Call

See `p1.c` for an example.

- Once the process reaches the `fork()` syscall, the OS will create a (nearly) identical copy of the calling process, just in its own address space
- From the perspective of the OS, there are two copies of `p1.c` running, each about to return from `fork()`.
- The child does not start at `main()`, but rather the site in which `fork()` was invoked.
- For the parent, `fork()` will return the PID of the child, and for the child it returns 0.
- The output is also not deterministic: either the child or parent may print first. This is determined by the CPU scheduler.

5.2 The `wait()` System Call

See `p2.c` for an example.

- In this example the parent invokes the `wait()` syscall, which blocks it until the child finishes executing.
- This will ensure that the program is now deterministic, as the child will always print first.

5.3 The `exec()` System Call

See `p3.c` for an example.

- On success, `exec()` doesn't return
- It will take the current process and replaces it with a different executable and starts at `main()`.

5.4 Motivating the API

- The separation between `fork()` and `exec()` allows the shell to run code after the `fork()` but before the `exec()`.
- This can alter the environment of the program that is to be executed.
- The shell shows a prompt and waits for the user to type a command.
- It then forks a new child process, calls `exec` to run the program, and waits for the command to finish, in which it prompts the user again.
- The separation between the fork and `exec` allows the shell to implement some useful behaviour, including:

```
1 prompt> wc p3.c > newfile.txt
```

- The `>` redirects the output of `wc p3.c` into `newfile.txt`. This is done by first forking a new child process and then opening `newfile.txt` before executing `wc p3.c`.
- This redirects the output from `stdout` to `newfile.txt`. See `p4.c` for an example.
- `close()` takes a file descriptor as a parameter, and deallocates it (for whichever process called it). Note that it does not stop/terminate any processes or free PIDs.

- The stdout fd is always 1 (defined via `STDOUT_FILENO` and is now free), and `open()` will always return the lowest fd, which is now set to `p4.output`.
- `wc p4.c` then writes to `p4.output` instead of stdout.

For the child process:				
	fd 0	fd 1	fd 2	
after <code>fork()</code>	tty	stdout	tty	(copied from parent)
after <code>close(1)</code>	tty	(free)	tty	
after <code>open(...)</code>	tty	<code>p4.output</code>	tty	<code>open</code> returns lowest fd = 1

- A similar implementation is done for `pipe()`, which lets you connect the output a process to the kernel pipe (or a queue), which connects to the input of another process. For example,

```
1 prompt> grep -o foo file | wc -l
```

- Will get the number of times `foo` appears in `file` and pipe it to `wc`.

5.5 Process Control and Users

- `kill()` is used to send signals to a process, including directives to pause, die and other things.
- On most shells, keystrokes can also send signals (`ctrl-c` sends `SIGINT`, graceful termination and `ctrl-z` sends `SIGSTP`, which pauses a program and can be resumed with something like `fg`.)
- A process should also implement the `signal()` syscall to catch different signals; for example, to cleanup resources when a `SIGINT` is issued.
- Modern systems require a strong notion of a "user" to ensure that different people have various levels of access to system resources and signals.

6 Mechanism: Limited Direct Execution

How to efficiently virtualize the CPU with control: The OS must virtualize the CPU in an efficient manner while retaining control over the system. Both hardware and OS support is required.

6.1 Basic Technique: Limited Direct Execution

- A technique used by operating systems to run programs on the CPU while keeping control of system security and resources.
- "Direct execution" refers to the fact that programs run on the CPU

OS	Program
Create an entry for process list	
Allocate memory for program	
Load program into memory	
Setup stack with argc and argv	
Clear registers	
Execute main()	Run main()
	Execute return from main()
Free memory of process	
Remove from process list	

Table 1: Direct execution protol, no limits

- Issues: How do we restrict operations for specific programs? How do we implement time sharing to virtualize the CPU?

6.2 Problem #1: Restricted Operations

How to perform restricted operations: A process must be able to perform I/O and some other (possibly) restricted operations, but without giving the process full access over the system. How does the OS implement this while still executing as fast as possible?

Definition 6.1 (User mode). User mode is a restricted operating state. Code ran in user mode does not have access to certain operations, such as issuing I/O requests. Performing a restricted operation in user mode will cause the processor to raise an exception and the OS to kill the proces.

Definition 6.2 (Kernel mode). The operating state in which the operating system (or kernel) runs in. There are no hardware-enforced restrictions in this mode.

- System calls allow the kernel to expose pieces of kernel functionality to user programs, such as accessing file systems, creating and destroying processes, communicating with other processes and allocating memory.
- System calls are executed via trap instructions, which is a jump + elevation into kernel mode.
- When finished, the OS issues a return-from-trap instruction, which returns back to the user program and reduces priviledges to user mode.
- The kernel controls what code executes during a trap with a trap table (set on boot).

OS @ boot (kernel mode)	Hardware	
Initialize trap table	remember address of syscall handler	
OS @ run (kernel mode)	Hardware	Program (user mode)
Create entry for process list		
Allocate memory for program		
Load program into memory		
Setup user stack with argv		
Fill kernel stack with reg/PC		
return-from-trap	restore regs(from kernel stack) move to user mode jump to main	Run main() /*executing..*/ Call syscall trap into OS
	save regs (to kernel stack) move to kernel mode jump to trap handler (from trap table)	
Handle trap		
Do work for syscall		
return-from-trap	restore regs(from kernel stack) move to user mode jump to PC	/*keep executing...*/ return from main trap invoked by exit()
Free memory of process		
Remove from process list		

Table 2: Limited Direct Execution Protocol

- The trap table contains various exceptions and one entry for syscalls (eg 0x80 on x86).
- When a syscall is invoked, a system-call number is assigned to each unique syscall. This tells the trap handler which syscall to invoke.
- User trap instruction -> hardware trap table -> syscall trap handler -> syscall table (seperate from trap table) -> do stuff

6.3 Problem #2: Switching Between Processes

Assuming a single core system: If a process is running on the CPU, then the OS is not running; how does it do anything?

Specifically, How can the operating system regain control of the CPU so that it can switch between processes?

6.3.1 Cooperative approach: Wait for syscalls

- Most early operating systems
- A cooperative approach means that the OS will assume that long-running processes will periodically give up the CPU so that the OS can do some other stuff.
- This works as most processes frequently issue syscalls to the OS.

- Cooperative operating systems also generally come with a yield syscall which transfers control to the OS so it can do stuff.
- Illegal operations (like dividing by 0, accessing restricted memory) will also generate traps for the OS so that it has control.
- However, this raises the question: How can the OS gain control of the CPU even if processes are not being cooperative?

6.4 Non-cooperative approach

- Timer interrupt: An internal timer will make the hardware periodically issue interrupts to a process every few milliseconds; this will temporarily halt the process and give the OS control through a trap (via an interrupt handler).
- The hardware also has some responsibility during an interrupt: it needs to save enough of the state of the program just before the interrupt so that after the subsequent return-from-trap instruction the program can run correctly.

6.4.1 Saving and Restoring Context

- The scheduler may decide to switch to a different process during an interrupt.
- If this happens, the OS runs a context switch, which saves GPRs, the PC, and the kernel stack pointer for the current process and restores them for the process to be executed.

OS @ boot (kernel mode)	Hardware	
initialize trap table		
	Remember address of syscall handler and timer handler	
start interrupt timer	start timer interrupt CPU in X ms	
OS @ run (kernel mode)	Hardware	Program(user mode)
		Process A /* after X ms... */
	timer interrupt regs(A)->kernel stack(A) move to kernel mode jump to trap handler	
Handle the trap Context switch: regs(A)->proc_t(A) regs(B)<-proc_t(B) switch to kernel stack(B) return from trap to B		
	regs(B)<-kernel stack(B) move to user mode jump to B's PC	
		Process B

tableLimited Direct Execution Protol (Timer Interrupt)